University of Cape Town

Honours Analytics 2026

Continuous Assessment 3

Start: 5 May 2026 8pm
End: 15 May 2026 23:59pm

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MS Teams

May 5, 2026

Instructions

1. You will need to submit 2 files in the Google/One Drive:
   - Your **pdf** file named with your student number which includes (i) a cover page, (ii) plagiarism declaration and (iii) your write-up of the analysis/results you have done. You can prepare this pdf file using any typesetting software you wish, MS Word, R Markdown, LaTeX etc. However, I would encourage you to use LaTeX (there is a template for this on Vula). This pdf document SHOULD NOT HAVE any R CODE or R OUTPUT! Only describe what you have done and what your results are. You can include R output and R graphs in a formatted/tabulated way, ie. figures and tables, not directly copy-pasted from R console! Do not give any plots that you have not talked about within the text. Plots are for visually summarizing your findings, no point in just giving a plot for the sake of it. Make sense of it and explain it. You cannot expect the reader to get their own understanding from the plots, this is your report not mine or someone else's. Provide Table and Figure captions and refer to these within your text.
   - Your R code named with your student number as an R script or an R markdown file. Do not forget to set the seed `set.seed(123)` if you are using a random starting point.
2. Hand-in date and time is 15 May 2026 23:59pm (negotiable).
3. I am available for consultation during the period of the CA between 10am-6pm. You can contact me via email or MS Teams messages and we can make MS Teams meetings individually. Please do not use the forum. Directly contact me.
4. Any communication between students will be subject to regular disciplinary processes and can lead to expulsion.

Problem Description and Expectations

There is one dataset saved as an .RData file `airbnbListings.RData` which originally was generated from the (cpt_listings) data that is downloaded from Inside Airbnb website.

Load the data using the following:

```r
load("airbnbListings.RData")
```

Use `class()` function to check the class of the dataset. The dataframe for this assignment includes several variables for listings that appear in Airbnb website for City of Cape Town (obtained from Bader 1919's Github page). The description of the variables scraped can be found from from Inside Airbnb website.

Most Airbnb datasets have:

- Property characteristics: Accommodates, bedrooms, bathrooms, beds, property type, room type
- Pricing & demand: Price, availability, number of reviews, review scores
- Host behaviour: Host listings count, superhost status
- Location: Latitude, longitude, Neighbourhood / ward

Part 1: Exploratory Data Analysis - 5%

Any analysis starts with knowing your data, therefore in this part of the assignment, you are expected to check all your variables first, use `str()` to see the types of your variables. You are also expected to detect any discrepancies/anomalies/missing values in your dataset using any or all of the following: tables, summaries, frequencies, plots. If need be, you may apply some dimension reduction techniques.

Some useful data preprocessing:

- clean and variable mutate
- dummy variables
- data overview
- ward level averages

Part 2: Cluster Analysis - 25% (HC) + 35% (NHC)

In the second part of the assignment, you will use some of the variables in order to apply one hierarchical (HC) and none-hierarchical clustering (NHC) algorithms. I expect to see comparative analysis. Be careful in the choice of your variables, you do not need to use every variable here. Report your findings.

Part 3: Self Organising Map - 25%

In the last part of the assignment, you will apply self-organising maps and in order to do so you may choose the size(s) of the grid(s) yourself, use the recommended approaches. How well do the clusters that you obtained in SOMs compare to the previous results? Report your findings.

Overall Presentation - 10%

You are expected to provide a short description of the dataset and the variables; a short description of any transformations, standardization, data cleaning applied; a very short description of each technique; a comparison of their performances where possible. I expect a neat report, so please try to be neat and clear.